Question			Answer	Marks	AO	Guidance
3	(a)		$\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}=\frac{25}{12}$	B1 [1]	1.1	Must be in lowest terms. Allow $2\frac{1}{12}$ BC
3	(b)		Sum $=\sum_{r=1}^{5}(2r-1)$ or $\sum_{r=0}^{4}(2r+1)$	M1 A1 [2]	3.1a 2.5	Correct function to generate odd numbers. Condone no brackets Correct limits for their expression using sigma notation Brackets must be seen. Do not allow $\sum_{r=1}^{5}(r+(r-1))$, $\sum_{r=1}^{5}(2n-1)$ or similar
3	(c)		The series is increasing, so the sum of *n* terms forms a divergent sequence so there is not well defined sum to infinity	B1 [1]	2.4	Explains in technical language or otherwise that the sum gets bigger the more terms added in so the sum to infinity is not a number. See appendix

Exemplar responses for Q3(c)

Response	Mark
As it is a divergent sequence [and ever increases]	**B1**
The sequence is not convergent	**B1**
The nth term value is constantly increasing	**B1**
It is constantly increasing	**B0**
It could go on indefinitely	**B0**
Because we can't sum to infinity arithmetic sequences	**B0**
It is an arithmetic sequence	**B0**
The sequence is not geometric sequence so the sum to infinity is infinity	**B0**